Unit 2

Structural Organisation in Plants and Animals

The diverse forms of life on earth were described only by observation, through naked eyes or later through lenses and microscopes, mainly as gross structural features.

When biology was natural history

- observable living phenomena were also recorded as part of this description
- before experimental biology, or physiology, was established as part of biology, naturalists described only biology
- so biology remained a natural history for a long time, and the description by itself was amazing in its detail

Why the detail mattered

- a student's initial reaction could be boredom, but the detailed description was used in later day reductionist biology, where living processes drew more attention than the description of life forms
- the description thus became meaningful and helpful in framing research questions in physiology or evolutionary biology

What this unit covers

- the structural organisation of plants and animals, including the structural basis of physiological or behavioural phenomena
- for convenience it is presented separately for plants and for animals

In this unit

Ch 5 Morphology of Flowering Plants

Ch 6 Anatomy of Flowering Plants

Ch 7 Structural Organisation in Animals

Katherine Esau, 1898 to 1997

- born in Ukraine in 1898
- studied agriculture in Russia and Germany, and received her doctorate in 1931 in the United States
- her early publications reported that the curly top virus spreads through a plant via the food-conducting or phloem tissue

Two books that changed the field

- Plant Anatomy, published in 1954, took a dynamic developmental approach designed to enhance understanding of plant structure, and had an enormous impact worldwide, bringing about a revival of the discipline
- The Anatomy of Seed Plants followed in 1960, and was referred to as the Webster's of plant biology, being encyclopediac

Recognition

- elected to the National Academy of Sciences in 1957, becoming the sixth woman to receive that honour
- received the National Medal of Science from President George Bush in 1989
- when she died in 1997, Peter Raven, director of Anatomy and Morphology at the Missouri Botanical Garden, remembered that she absolutely dominated the field of plant biology even at the age of 99

One body plan behind the variety

- the wide range in the structure of higher plants never fails to fascinate
- even though angiosperms show such large diversity in external structure or morphology, they are all characterised by the presence of roots, stems, leaves, flowers and fruits

Why the technical terms matter

- for any successful attempt at classification, and at understanding any higher plant, we need standard technical terms and standard definitions
- we also need to know the possible variations in different parts, found as adaptations to the environment: to various habitats, for protection, climbing and storage

Two systems

- pull out any weed and it will have roots, stems and leaves, and may bear flowers and fruits
- the underground part of the flowering plant is the root system
- the portion above the ground forms the shoot system

5.1 The root

The primary root

- in the majority of dicotyledonous plants the direct elongation of the radicle forms the primary root, which grows inside the soil
- it bears lateral roots of several orders, referred to as secondary, tertiary and so on

Shoot system above ground, with stem, leaf, bud, flower and fruit

Node where a leaf is borne, internode the length between two nodes

Root system below ground, with primary root and secondary root

Figure 5.A Parts of a flowering plant

Tap, a main root with laterals, as in mustard

Fibrous, many roots from the stem base, as in wheat

Adventitious, roots from parts other than the radicle

Figure 5.B Different types of roots : (a) Tap (b) Fibrous (c) Adventitious

Three kinds of root system

- the primary root and its branches constitute the tap root system, as seen in the mustard plant
- in monocotyledonous plants the primary root is short lived and is replaced by a large number of roots originating from the base of the stem, the fibrous root system, as in wheat
- in some plants such as grass, Monstera and the banyan tree, roots arise from parts of the plant other than the radicle and are called adventitious roots

What the root system does

- absorption of water and minerals from the soil
- providing proper anchorage to the plant parts
- storing reserve food material
- synthesis of plant growth regulators

Root cap, the thimble-like cover at the apex

Region of meristematic activity, small thin-walled cells that divide repeatedly

Region of elongation, where cells enlarge and lengthen the root

Region of maturation, where cells differentiate and root hairs arise

Figure 5.C The regions of the root-tip

5.2 Regions of the root

From cap to root hair

- the root cap protects the tender apex as it makes its way through the soil
- a few millimetres above it is the region of meristematic activity, whose cells are very small, thin-walled, with dense protoplasm, and divide repeatedly
- cells proximal to that region undergo rapid elongation and enlargement, and are responsible for growth of the root in length: the region of elongation
- proximal again, cells gradually differentiate and mature in the region of maturation, where some epidermal cells form fine delicate thread-like root hairs that absorb water and minerals

5.3 The stem

What distinguishes a stem

- the ascending part of the axis, bearing branches, leaves, flowers and fruits
- it develops from the plumule of the embryo of a germinating seed
- it bears nodes, where leaves are born, and internodes, the portions between two nodes
- it bears buds, which may be terminal or axillary, and is generally green when young, later often becoming woody and dark brown

What the stem does

- spreading out branches bearing leaves, flowers and fruits
- conducting water, minerals and photosynthates
- some stems store food, or give support, protection and vegetative propagation

5.4 The leaf

A lateral outgrowth of the stem

- generally flattened, borne on the stem, developing at the node and bearing a bud in its axil, which later develops into a branch
- leaves originate from shoot apical meristems and are arranged in an acropetal order
- they are the most important vegetative organs for photosynthesis

Leaf base attaching the leaf to the stem, sometimes with stipules

Petiole, the stalk that holds the blade to light

Lamina, the green expanded blade with veins and veinlets

Reticulate venation, a network; parallel venation, veins running alongside

Figure 5.D Structure of a leaf : (a) Parts of a leaf (b) Reticulate venation (c) Parallel venation

Pinnately compound, leaflets along a common axis, the rachis

Palmately compound, leaflets meeting at the tip of the petiole

Figure 5.E Compound leaves : (a) pinnately compound leaf (b) palmately compound leaf

Three main parts

- leaf base, petiole and lamina
- the leaf attaches to the stem by the leaf base, which may bear two lateral small leaf like structures called stipules
- in monocotyledons the leaf base expands into a sheath covering the stem partially or wholly, and in some leguminous plants it swells into a pulvinus
- long thin flexible petioles let blades flutter in wind, cooling the leaf and bringing fresh air to its surface

The lamina

- the green expanded part of the leaf, with veins and veinlets, and usually a prominent middle vein, the midrib
- veins give rigidity to the blade and act as channels of transport for water, minerals and food materials
- shape, margin, apex, surface and extent of incision of the lamina vary in different leaves

5.5 Venation and types of leaves

Venation

- the arrangement of veins and veinlets in the lamina
- reticulate when the veinlets form a network, generally in dicotyledonous plants
- parallel when the veins run parallel to each other within a lamina, characteristic of most monocotyledons

Simple and compound

- a leaf is simple when its lamina is entire, or when incised, the incisions do not touch the midrib
- when the incisions reach the midrib and break it into leaflets, the leaf is compound
- a bud is present in the axil of the petiole in both simple and compound leaves, but not in the axil of the leaflets of a compound leaf

Alternate, a single leaf at each node

Opposite, a pair at each node lying opposite one another

Whorled, more than two leaves at a node forming a whorl

Figure 5.F Different types of phyllotaxy : (a) Alternate (b) Opposite (c) Whorled

The main axis continues to grow

Flowers borne laterally in acropetal succession

Figure 5.G Racemose inflorescence

The two compound types

- in a pinnately compound leaf a number of leaflets are present on a common axis, the rachis, which represents the midrib of the leaf, as in neem
- in palmately compound leaves the leaflets are attached at a common point, at the tip of the petiole, as in silk cotton

5.6 Phyllotaxy

Arrangement of leaves on the stem

- alternate, a single leaf arising at each node in alternate manner, as in china rose, mustard and sun flower
- opposite, a pair arising at each node lying opposite each other, as in Calotropis and guava
- whorled, more than two leaves arising at a node to form a whorl, as in Alstonia

5.7 The inflorescence

A flower is a modified shoot

- the shoot apical meristem changes to a floral meristem, internodes do not elongate and the axis gets condensed
- the apex produces different kinds of floral appendages laterally at successive nodes instead of leaves, and when a shoot tip transforms into a flower it is always solitary
- the arrangement of flowers on the floral axis is the inflorescence

Racemose and cymose

- the two major types are defined by whether the apex develops into a flower or continues to grow
- in racemose types the main axis continues to grow and the flowers are borne laterally in acropetal succession
- in cymose types the main axis terminates in a flower and so is limited in growth, the flowers borne in basipetal order

The main axis ends in a flower and growth is limited

Flowers borne in basipetal order

Figure 5.H Cymose inflorescence

Hypogynous, gynoecium highest, ovary superior

Perigynous, other parts on the thalamus rim, ovary half inferior

Epigynous, thalamus encloses the ovary, ovary inferior

Figure 5.I Position of floral parts on thalamus : (a) Hypogynous (b) and (c) Perigynous (d) Epigynous

5.8 The flower

Four whorls on the thalamus

- the flower is the reproductive unit in angiosperms, meant for sexual reproduction
- a typical flower has four kinds of whorls arranged successively on the swollen end of the stalk or pedicel, called the thalamus or receptacle
- these are calyx, corolla, androecium and gynoecium; calyx and corolla are accessory organs while androecium and gynoecium are reproductive organs
- in some flowers such as lily the calyx and corolla are not distinct and are termed perianth

Sex and symmetry

- a flower with both androecium and gynoecium is bisexual; one with only stamens or only carpels is unisexual
- actinomorphic, or radially symmetrical, when it can be divided into two equal radial halves in any radial plane through the centre, as in mustard, datura and chilli
- zygomorphic, or bilaterally symmetrical, when it divides into two similar halves in only one particular vertical plane, as in pea, gulmohur, bean and Cassia
- asymmetric or irregular if no vertical plane through the centre divides it into similar halves, as in canna

Merosity and bracts

- a flower is trimerous, tetramerous or pentamerous when the floral appendages are in multiples of 3, 4 or 5
- flowers with bracts, the reduced leaf found at the base of the pedicel, are bracteate
- those without bracts are ebracteate

Calyx, the outermost whorl

Corolla, within the calyx

Androecium, the male whorl

Gynoecium, at the centre

Pedicel, the stalk below the thalamus

Figure 5.K Parts of a flower

5.9 Position of the parts on the thalamus

Three positions

- in a hypogynous flower the gynoecium occupies the highest position and the other parts sit below it, so the ovary is superior, as in mustard, china rose and brinjal
- if the gynoecium is central and the other parts lie on the rim of the thalamus at almost the same level, the flower is perigynous and the ovary half inferior, as in plum, rose and peach
- in epigynous flowers the margin of the thalamus grows upward enclosing the ovary completely and fusing with it, the other parts arising above, so the ovary is inferior, as in guava, cucumber and the ray florets of sunflower

5.10 Calyx and corolla

Calyx

- the outermost whorl, its members called sepals
- sepals are generally green and leaf like, and protect the flower in the bud stage
- the calyx may be gamosepalous, with sepals united, or polysepalous, with sepals free

Corolla

- composed of petals, usually brightly coloured to attract insects for pollination
- like the calyx it may be gamopetalous, petals united, or polypetalous, petals free
- shape and colour vary greatly; corolla may be tubular, bell-shaped, funnel-shaped or wheel-shaped

Valvate, margins just touching without overlap

Twisted, one margin overlapping the next

Imbricate, margins overlapping in no particular direction

Vexillary, standard over wings over keel

Figure 5.L Types of aestivation in corolla : (a) Valvate (b) Twisted (c) Imbricate (d) Vexillary

5.11 Aestivation

Four modes

- aestivation is the mode of arrangement of sepals or petals in the floral bud with respect to the other members of the same whorl
- valvate, when they just touch one another at the margin without overlapping, as in Calotropis
- twisted, when one margin overlaps that of the next and so on, as in china rose, lady's finger and cotton
- imbricate, when margins overlap one another but not in any particular direction, as in Cassia and gulmohur

Vexillary

- in pea and bean flowers there are five petals
- the largest, the standard, overlaps the two lateral petals, the wings
- the wings in turn overlap the two smallest anterior petals, the keel
- this type is known as vexillary or papilionaceous

5.12 Androecium

Stamens

- the androecium is composed of stamens, each representing the male reproductive organ and consisting of a stalk or filament and an anther
- each anther is usually bilobed and each lobe has two chambers, the pollen-sacs, in which pollen grains are produced
- a sterile stamen is called a staminode

How stamens unite

- attached to the petals they are epipetalous, as in brinjal; attached to the perianth they are epiphyllous, as in lily
- free stamens are polyandrous; united they may form one bundle, monoadelphous, as in china rose, two bundles, diadelphous, as in pea, or more than two, polyadelphous, as in citrus
- filament length may vary within a flower, as in Salvia and mustard

Marginal, ovules in two rows on a ridge along the ventral suture

Axile, ovules on an axial placenta in a multilocular ovary

Parietal, ovules on the inner wall or peripheral part

Free central, ovules on a central axis with no septa

Basal, placenta at the base with a single ovule

Figure 5.M Types of placentation : (a) Marginal (b) Axile (c) Parietal (d) Free central (e) Basal

5.13 Gynoecium

The carpel

- the female reproductive part, made up of one or more carpels, each of stigma, style and ovary
- the ovary is the enlarged basal part, on which lies the elongated tube, the style, which connects the ovary to the stigma
- the stigma is usually at the tip of the style and is the receptive surface for pollen grains
- each ovary bears one or more ovules attached to a flattened cushion-like placenta

Free or fused

- when more than one carpel is present they may be free, as in lotus and rose, and are then apocarpous
- they are syncarpous when the carpels are fused, as in mustard and tomato
- after fertilisation the ovules develop into seeds and the ovary matures into a fruit

5.14 Placentation

Where the ovules sit

- placentation is the arrangement of ovules within the ovary: marginal, axile, parietal, basal, central and free central
- marginal, the placenta forming a ridge along the ventral suture with ovules in two rows, as in pea; axile, the placenta axial with ovules attached in a multilocular ovary, as in china rose, tomato and lemon
- parietal, ovules developing on the inner wall or peripheral part, the one-chambered ovary becoming two-chambered by a false septum, as in mustard and Argemone
- free central, ovules on a central axis with septa absent, as in Dianthus and Primrose; basal, the placenta at the base of the ovary with a single ovule, as in sunflower and marigold

5.15 The fruit

A ripened ovary

- the fruit is a characteristic feature of flowering plants, a mature or ripened ovary developed after fertilisation
- a fruit formed without fertilisation of the ovary is parthenocarpic
- generally the fruit consists of a wall or pericarp and seeds; the pericarp may be dry or fleshy, and when thick and fleshy it differentiates into outer epicarp, middle mesocarp and inner endocarp

Mango, epicarp thin, mesocarp fleshy and edible, endocarp stony

Coconut, also a drupe, with a fibrous mesocarp

Figure 5.N Parts of a fruit : (a) Mango (b) Coconut

Seed coat, of outer testa and inner tegmen

Hilum, the scar of attachment; micropyle, the pore above it

Cotyledons, often fleshy and full of reserve food

Radicle and plumule at the two ends of the embryonal axis

Figure 5.O Structure of dicotyledonous seed

Drupes

- in mango and coconut the fruit is a drupe, developing from monocarpellary superior ovaries and one seeded
- in mango the pericarp is well differentiated into an outer thin epicarp, a middle fleshy edible mesocarp and an inner stony hard endocarp
- in coconut, also a drupe, the mesocarp is fibrous

5.16 The seed

What a seed is

- the ovules after fertilisation develop into seeds
- a seed is made up of a seed coat and an embryo
- the embryo is made up of a radicle, an embryonal axis and one cotyledon, as in wheat and maize, or two, as in gram and pea

The dicotyledonous seed

- the outermost covering is the seed coat, of two layers, the outer testa and inner tegmen
- the hilum is a scar on the seed coat through which the developing seed was attached to the fruit, and above it is a small pore, the micropyle
- within the seed coat is the embryo, of an embryonal axis and two cotyledons, often fleshy and full of reserve food materials
- in seeds such as castor the endosperm formed by double fertilisation stores food, and these are endospermic; in bean, gram and pea the endosperm is absent at maturity and the seeds are non-endospermous

Seed coat and fruit-wall fused and membranous

Endosperm, bulky, storing food, with an aleurone layer outside

Scutellum, the single large shield shaped cotyledon

Plumule in the coleoptile, radicle in the coleorhiza

Figure 5.P Structure of a monocotyledonous seed

Calyx outermost, then corolla, androecium, gynoecium at the centre

A dot at the top marks the position of the mother axis

The formula records number, fusion and adhesion of the parts

Figure 5.R Floral diagram with floral formula

5.17 The monocotyledonous seed

Endospermic, mostly

- generally monocotyledonous seeds are endospermic, but some, as in orchids, are non-endospermic
- in cereals such as maize the seed coat is membranous and generally fused with the fruit wall
- the endosperm is bulky and stores food, its outer covering separated from the embryo by a proteinous aleurone layer
- the embryo is small and sits in a groove at one end of the endosperm, with one large shield shaped cotyledon, the scutellum, and a short axis carrying plumule and radicle enclosed in sheaths called coleoptile and coleorhiza

5.18 Semi-technical description

How a plant is described

- the description has to be brief, in simple scientific language, and presented in a proper sequence
- it begins with habit, then vegetative characters of root, stem and leaves, then floral characters of inflorescence and flower parts
- after the parts are described, a floral diagram and a floral formula are presented

The symbols

- Br stands for bracteate, K for calyx, C for corolla, P for perianth, A for androecium and G for gynoecium, with G marked for superior and for inferior ovary
- there are symbols for male, for female and for bisexual plants
- the symbol for actinomorphic flowers is given, and another for zygomorphic ones

Flowering twig

Flower

Longitudinal section of the flower

Stamens

Carpel

Floral diagram

Figure 5.S Solanum nigrum (makoi) plant : (a) Flowering twig (b) Flower (c) L.S. of flower (d) Stamens (e) Carpel (f) Floral diagram

Reading a floral formula and diagram

- fusion is indicated by enclosing the figure within a bracket, and adhesion by a line drawn above the symbols of the floral parts
- a floral diagram gives the number of parts of a flower, their arrangement and their relation to one another
- the position of the mother axis with respect to the flower is shown by a dot on the top of the diagram, and calyx, corolla, androecium and gynoecium are drawn in successive whorls, calyx outermost and gynoecium at the centre
- the floral formula also shows cohesion and adhesion within whorls and between whorls, and the example given represents the mustard plant, of family Brassicaceae

5.19 Solanaceae

The potato family

- a large family, commonly called the potato family
- widely distributed in tropics, subtropics and even temperate zones

Vegetative characters

- plants mostly herbs, shrubs and rarely small trees
- stem herbaceous, rarely woody, aerial, erect, cylindrical, branched, solid or hollow, hairy or glabrous, with an underground stem in potato, Solanum tuberosum
- leaves alternate, simple, rarely pinnately compound, exstipulate, with reticulate venation

Floral characters

- inflorescence solitary, axillary or cymose as in Solanum; flower bisexual and actinomorphic
- calyx of five sepals, united, persistent, with valvate aestivation; corolla of five petals, united, with valvate aestivation
- androecium of five stamens, epipetalous; gynoecium bicarpellary, obligately placed, syncarpous, the ovary superior and bilocular with a swollen axile placenta bearing many ovules
- fruits a berry or capsule; seeds many and endospermous, and the floral formula is written with the symbols already given

Economic importance

- food, from tomato, brinjal and potato
- spice, from chilli
- medicine, from belladonna and ashwagandha
- fumigatory from tobacco, and ornamentals such as petunia

5.20 Summary

Root and shoot

- flowering plants show enormous variation in shape, size, structure, mode of nutrition, life span, habit and habitat, with well developed root and shoot systems
- the root system is either tap or fibrous, dicotyledonous plants generally having tap roots and monocotyledonous plants fibrous roots; roots in some plants are modified for storage of food, mechanical support and respiration
- the shoot system is differentiated into stem, leaves, flowers and fruits, and nodes and internodes, multicellular hair and positively phototropic nature distinguish stems from roots
- the leaf is a lateral outgrowth of the stem developed exogeneously at the node, green to perform photosynthesis, and varies markedly in shape, size, margin, apex and incision

The flower, and why description matters

- the flower is a modified shoot meant for sexual reproduction, and flowers are arranged in different types of inflorescences
- they vary enormously in structure, symmetry, position of ovary in relation to other parts, and arrangement of petals, sepals and ovules
- after fertilisation the ovary is modified into fruit and the ovules into seeds, which may be monocotyledonous or dicotyledonous and vary in shape, size and period of viability
- floral characteristics form the basis of classification and identification of flowering plants, shown through semi-technical descriptions of families, with floral diagrams and floral formulae as the summarised form

5.21 Exercises

From this chapter

1. How is a pinnately compound leaf different from a palmately compound leaf?
2. Explain with suitable examples the different types of phyllotaxy.
3. Define these terms: (a) aestivation (b) placentation (c) actinomorphic (d) zygomorphic (e) superior ovary (f) perigynous flower (g) epipetalous stamen
4. Differentiate between (a) racemose and cymose inflorescence (b) apocarpous and syncarpous ovary
5. Draw the labelled diagram of the following: (i) gram seed (ii) V.S. of maize seed
6. Take one flower of the family Solanaceae and write its semi-technical description. Also draw its floral diagram.
7. Describe the various types of placentations found in flowering plants.
8. What is a flower? Describe the parts of a typical angiosperm flower.
9. Define the term inflorescence. Explain the basis for the different types of inflorescence in flowering plants.
10. Describe the arrangement of floral members in relation to their insertion on thalamus.